DS5006 Lab 3 — Report

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SQ1 — Why do we not split on the thing we actually care about? (6)

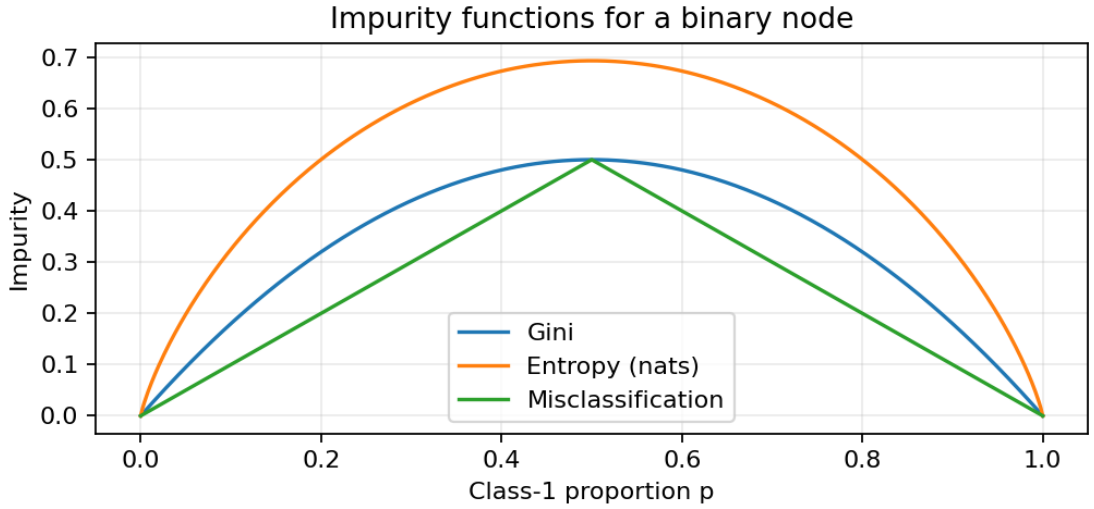
Best split at the tricky node

Criterion	Feature	Threshold	Gain
Gini	0	-1.044356	0.043393
Entropy	0	-1.044356	0.055506
Misclassification	-1 (no split)	NaN	0.000000

Trees grown with each criterion

Depth	Gini err	Gini leaves	Entropy err	Entropy leaves	Misclass err	Misclass leaves
1	0.3473	2	0.3473	2	0.3667	2
2	0.1627	4	0.1627	4	0.2467	4
3	0.1173	8	0.1083	8	0.1350	8
4	0.1173	13	0.1103	15	0.1250	13
5	0.1313	23	0.1243	27	0.1350	16
6	0.1397	36	0.1323	42	0.1307	18
8	0.1707	65	0.1663	73	0.1307	18
12	0.2033	104	0.1987	103	0.1307	18
20	0.2070	106	0.1987	103	0.1307	18

What I infer. At the tricky node, Gini and entropy select feature 0 at threshold -1.044356 and achieve positive gains, whereas misclassification impurity returns no admissible split. This is not simply a case where misclassification chooses a worse split: its gain is exactly zero for every candidate split at this node under the gain tolerance. For binary class proportion p, misclassification impurity M(p)=min(p,1-p) is piecewise linear with a kink at p=0.5. Splits that change class proportions without changing the majority class can therefore produce no reduction in this impurity. Gini, G(p)=2p(1-p), and entropy, H(p)=-p log p-(1-p) log(1-p), are curved and can reward such changes. The depth sweep also shows that test error does not improve monotonically with depth. Entropy gives the lowest error at depth 3 (0.1083), while misclassification remains at error 0.1307 at depths 6–20.



SQ2 — How deep, and how much should you trust it? (8)

The best depth by the Gini test-error sweep is **3**, with test error **0.117333**. At depth 20, test error is **0.207000**, a gap of **0.089667**. This is evidence that growing the tree deeper can overfit: a more complex tree can fit training details that do not generalize to held-out data.

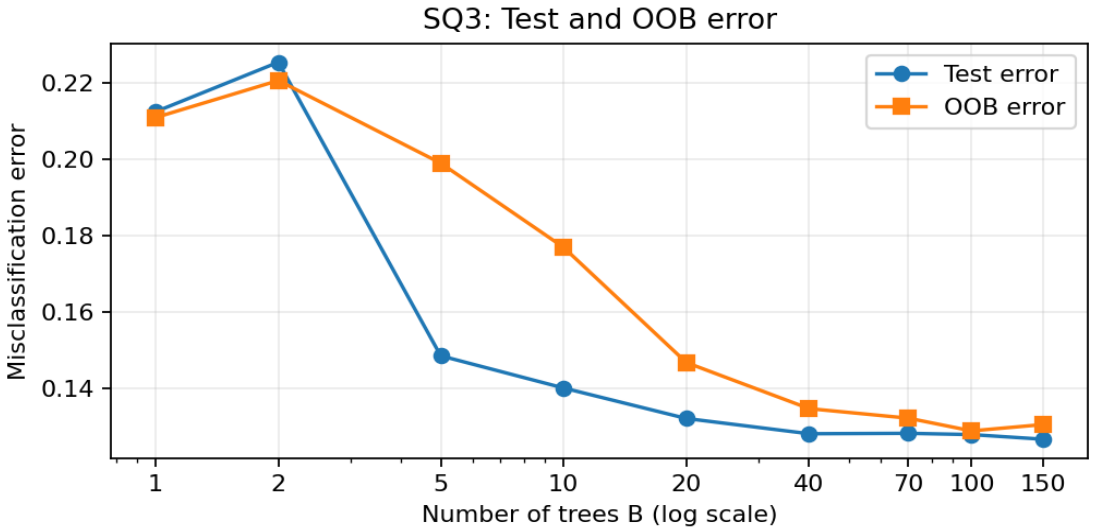
Stability over 15 bootstrap resamples. Root feature counts for fully grown trees were: feature 0 = 0, feature 1 = 0, feature 2 = 15, feature 3 = 0, feature 4 = 0, feature 5 = 0. All 15 fully grown trees selected feature 2 at the root.

Measure	Fully grown	At best depth (3)
Mean pairwise disagreement	0.190667	0.069784

The minimum and maximum individual test errors among fully grown trees were **0.192000 to 0.227667**. Because the fully grown and depth-3 trees were trained on the same bootstrap resamples, the lower disagreement among depth-3 trees suggests that much of the instability accumulates in deeper parts of the trees rather than at the root. The root counts reinforce this for this run: all 15 fully grown trees chose feature 2, so root-feature selection was stable even though the complete trees disagreed on about 19.1% of test predictions on average. Pruning at depth 3 reduced disagreement to about 7.0%. The shared root does not imply that subsequent splits were stable.

SQ3 — Bagging, and what “out of bag” actually estimates (8)

Figure 1: test error and OOB error against B, averaged over three forests.



Single tree at BAG_DEPTH on the full training set: **0.207000**. Mean test error of individual bagged trees: **0.212981**. Test error of the first four trees: **0.176444**.

B	Test error	OOB error
1	0.212444	0.210928
2	0.225444	0.220639
5	0.148444	0.198973
10	0.140000	0.176891
20	0.132000	0.146667
40	0.128000	0.134583
70	0.128111	0.132083
100	0.127778	0.128750
150	0.126556	0.130417

Saturation. The test-error curve begins to flatten at approximately 40 trees. It falls from 0.212444 at one tree to 0.126556 at 150 trees, a reduction of 0.085888 (8.59 percentage points). Between 40 and 150 trees it changes only from 0.128000 to 0.126556. Bagging reduces variation among trees but does not eliminate shared errors. The ensemble outperforms the full-data tree and the mean individual bagged tree. The best depth-3 tree from SQ2 has error 0.117333, lower than the bagged result, although this comparison involves different tree depths.

SQ3 — Level and shape analysis

(a) Level at B = 150

OOB minus test error is 0.130417 - 0.126556 = **0.003861**.

SE(OOB) = sqrt[0.130417(1 - 0.130417)/800] ≈ 0.011906.

SE(test) = sqrt[0.126556(1 - 0.126556)/3000] ≈ 0.006070.

SE(difference) ≈ sqrt(0.011906² + 0.006070²) = **0.013364**. The absolute difference is 0.003861/0.013364 ≈ **0.289 standard errors**, small relative to estimated sampling noise. This run does not provide strong evidence that the OOB and test error levels differ. This is approximate because OOB predictions share trees and are dependent.

(b) Shape: excess above each curve's own plateau

	OOB excess	Test excess at same B	Test excess of 4 trees
B = 10	0.046475	0.013444	0.049889
B = 20	0.016250	0.005444	—

At B=10, the OOB excess (0.046475) differs from the test excess at B=4 (0.049889) by only 0.003414, compared with 0.033030 from the test excess at B=10 (0.013444). Each training observation is OOB for about 37% of trees, so at B=10 it receives about 3.7 OOB votes, roughly the four votes in the comparison. This supports the explanation that OOB excess decays more slowly because fewer trees vote on each training observation. At B=20 the OOB excess remains larger, consistent with slower convergence.

SQ4 — How strong does a weak learner have to be? (8)

Figure 2: training and test error against boosting round m, for base depths 1 and 3.



Base depth	m0 (train error hits 0)	Train error at M=300	Test error at M=300
1	Never	0.315000	0.354667
3	148	0.000000	0.150000

Why the two depths do not converge. A stump makes one threshold split. A sum of stumps can combine individual feature effects, but has a restricted additive form and may not efficiently represent complex feature interactions. A sum of depth-3 trees represents a richer family of decision functions because each tree makes several sequential splits. The target therefore appears to require structure that the depth-1 ensemble does not capture effectively: after 300 rounds its training error remains 0.315000, while depth 3 reaches zero training error at round 148. More rounds do not guarantee that a restricted base-learner family can close this representational gap.

The bound

At m=300, depth 3 has exponential-loss bound **0.03156**, training error **0.000000**, and test error **0.150000**. The bound was above training error at every round for both depths. It controls an upper bound on training classification error under the standard AdaBoost formulation; it does not bound test error or guarantee that test error decreases at every round.

Margins, depth 3, training set

Round	Minimum	5th percentile	Median
m0 = 148	0.0027	0.0396	0.1266
300	0.0327	0.0527	0.1059

What I infer. After round 148, every training point is correctly classified, but boosting continues to change the margin distribution. By round 300, the minimum margin and 5th percentile increased, while the median decreased. Thus, the weakest training example and lower tail gained margin, but the typical normalized margin did not increase. Since margins are normalized by the sum of absolute boosting weights, adding rounds changes both the cumulative score and its normalizer. Zero training error does not imply all margins continue to increase or that test error must improve.

Notes

26. Experiment scripts produced the required tables and plots.

Approximate total time spent: 28 hours

Declarations

1. Used AI to format the report document.
2. Referred to Pattern Recognition and Machine Learning by Christopher M. Bishop and Machine Learning: A Probabilistic Perspective by Kevin P. Murphy.
3. NumPy Documentation.